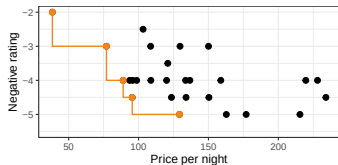
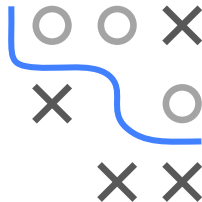


Optimization in Machine Learning

Multi-criteria Optimization Introduction



Learning goals

- What is multi-criteria optimization?
- Motivating examples
- Pareto optimality

FINDING PARETO-OPTIMAL SOLUTIONS I

- **Weighted sum scalarization** solves:

$$\mathbf{x}^* := \arg \min_{\mathbf{x} \in \mathcal{S}} \sum_{i=1}^m w_i f_i(\mathbf{x})$$

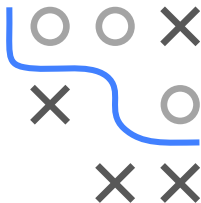
We can show that this approach can find Pareto-optimal solutions.

- Without loss of generality we can assume $\sum_{i=1}^d w_i = 1$.
- **Theorem:** If $w_i > 0$ for all i , then $\mathbf{x}^*$ is Pareto-optimal. **Proof.**

Suppose $\mathbf{x}^*$ is not Pareto-optimal, i.e., there exists $\mathbf{x}$ such that $f_i(\mathbf{x}) \leq f_i(\mathbf{x}^*)$ for all i and $f_j(\mathbf{x}) < f_j(\mathbf{x}^*)$ for at least one j . Then,

$$\sum_{i=1}^m w_i f_i(\mathbf{x}) < \sum_{i=1}^d w_i f_i(\mathbf{x}^*),$$

which contradicts the definition of $\mathbf{x}^*$ as the minimizer.

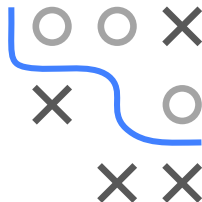
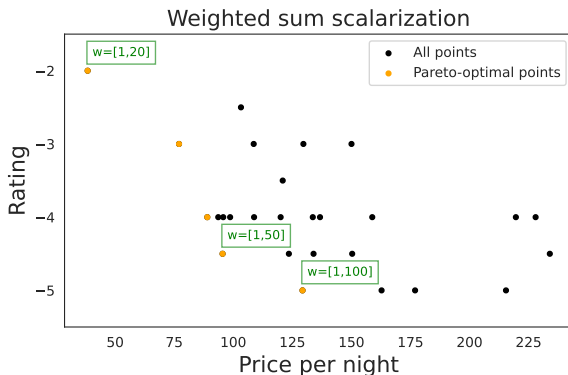


FINDING PARETO-OPTIMAL SOLUTIONS II

- Not all Pareto-optimal solutions can be found by solving

$$\mathbf{x}^* := \arg \min_{\mathbf{x} \in \mathcal{S}} \sum_{i=1}^m w_i f_i(\mathbf{x})$$

- Example:** We find three out of the five Pareto-optimal solutions:



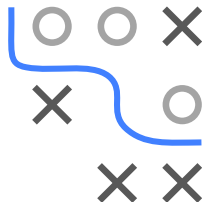
FINDING PARETO-OPTIMAL SOLUTIONS V

Theorem (Geoffrion, 1968): Let $\mathcal{S}$ be convex and assume $f_1, \dots, f_d$ are convex. Then $\mathbf{x}^* \in \mathcal{S}$ is properly Pareto-optimal if and only if $\mathbf{x}^*$ is an optimal solution of

$$\min_{\mathbf{x} \in \mathcal{S}} \sum_{i=1}^m w_i f_i(\mathbf{x})$$

with strictly positive weights w_i for all i .

- For convex functions and convex search spaces, we can find all properly Pareto-optimal solutions using weighted sum scalarization.
- **Proper Pareto-optimal solution:** Exclude solutions where an arbitrary small improvement in one objective leads to an arbitrary large deterioration in another objective.



FINDING PARETO-OPTIMAL SOLUTIONS VI

Definition (Geoffrion, 1968): A Pareto-optimal solution $\mathbf{x}^* \in \mathcal{S}$ is called **properly** Pareto-optimal, if there exists a constant $M > 0$ such that for all i and $\mathbf{x} \in \mathcal{S}$ with $f_i(\mathbf{x}) < f_i(\mathbf{x}^*)$ there exists an index j , such that $f_j(\mathbf{x}^*) < f_j(\mathbf{x})$ and

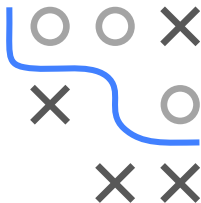
$$\frac{f_i(\mathbf{x}^*) - f_i(\mathbf{x})}{f_j(\mathbf{x}) - f_j(\mathbf{x}^*)} \leq M$$

Example: $f_1(x) = (x - 1)^2$ and $f_2(x) = 3(x - 2)^2$ for $0 \leq x \leq 3$.

- Pareto-optimal solutions are $x \in [1, 2]$ (as exercise).
- $x^* = 1$ is not properly Pareto-optimal.

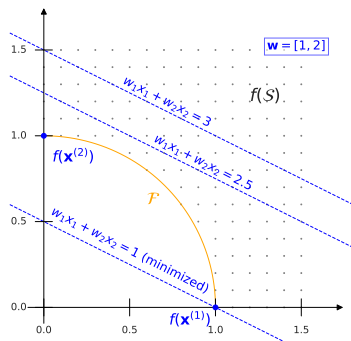
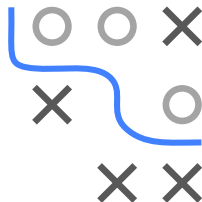
Proof. For $x^* = 1$ we have $f_1(x^*) = 0$ and $f_2(x^*) = 3$. Let $x = 1 + \epsilon$ for $\epsilon > 0$, then $f_2(x) < f_2(x^*)$ and $f_1(x) > f_1(x^*)$. Then,

$$\frac{f_2(x^*) - f_2(x)}{f_1(x) - f_1(x^*)} = \frac{3 - 3(\epsilon - 1)^2}{\epsilon^2} = \frac{6\epsilon - 3\epsilon^2}{\epsilon^2} \rightarrow \infty \text{ for } \epsilon \rightarrow 0$$



FINDING PARETO-OPTIMAL SOLUTIONS VII

- Convexity is required to find properly Pareto-optimal solutions
- In non-convex settings: Often poor coverage of the Pareto front.
- **Example:** $\mathcal{S} = \{\mathbf{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 \geq 1\}$ with $f_1(\mathbf{x}) = x_1$ and $f_2(\mathbf{x}) = x_2$.
- **Pareto front:** $\mathcal{F} = \{\mathbf{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\}$
- **Weighted sum scalarization** only finds $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$

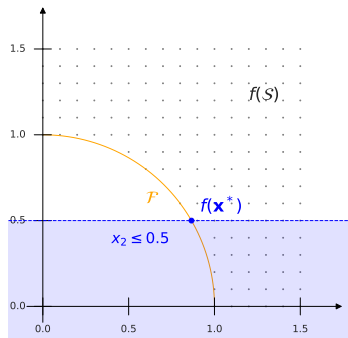
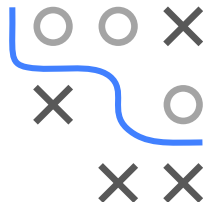


FINDING PARETO-OPTIMAL SOLUTIONS VIII

- **Alternative:** ϵ -constraint method

$$\mathbf{x}^* := \arg \min_{\mathbf{x} \in \mathcal{S}} f_1(\mathbf{x}) \quad \text{s.t.} \quad f_2(\mathbf{x}) \leq \epsilon_2.$$

- By varying ϵ_i we can find different Pareto-optimal solutions:



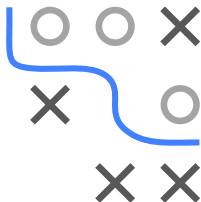
FINDING PARETO-OPTIMAL SOLUTIONS IX

- General form of the ϵ -constraint method:

$$\mathbf{x}^* := \arg \min_{\mathbf{x} \in \mathcal{S}} f_i(\mathbf{x}) \quad \text{s.t.} \quad f_j(\mathbf{x}) \leq \epsilon_j, \quad j = 1, \dots, m, j \neq i.$$

- Are solutions found by the ϵ -constraint method Pareto-optimal?
- **Theorem:** If $\mathbf{x}^*$ is the unique optimal solution of the ϵ -constraint method for some index i , then $\mathbf{x}^*$ is Pareto-optimal.

Proof. Suppose $\mathbf{x}^*$ is not Pareto-optimal. Then there exists $\mathbf{x} \in \mathcal{S}$, such that $f_i(\mathbf{x}) \leq f_i(\mathbf{x}^*)$ for all $i = 1, \dots, m$ and $f_j(\mathbf{x}) < f_j(\mathbf{x}^*)$ for at least one $j \in \{1, \dots, m\}$. Since $\mathbf{x}^*$ is feasible for the ϵ -constraint problem, it follows that $f_i(\mathbf{x}) \leq f_i(\mathbf{x}^*) \leq \epsilon_i$, so $\mathbf{x}$ is also feasible. Moreover, $f_1(\mathbf{x}) \leq f_1(\mathbf{x}^*)$, and since $\mathbf{x}^*$ is the unique minimizer of cost_1 , we must have $\mathbf{x} = \mathbf{x}^*$, which is a contradiction.



FINDING PARETO-OPTIMAL SOLUTIONS X

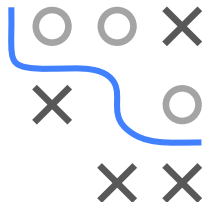
- Can the ϵ -constraint method find all Pareto-optimal solutions?

$$\mathbf{x}^* := \arg \min_{\mathbf{x} \in \mathcal{S}} f_i(\mathbf{x}) \quad \text{s.t. } f_j(\mathbf{x}) \leq \epsilon_j, \quad j = 1, \dots, m, j \neq i.$$

- **Theorem:** $\mathbf{x}^* \in \mathcal{S}$ is Pareto-optimal if and only if there exists $\epsilon \in \mathbb{R}^m$ such that $\mathbf{x}^*$ is an optimal solution for all $i = 1, \dots, m$.

Proof. “ $\Rightarrow$ ” Suppose $\mathbf{x}^*$ is not an optimal solution of the ϵ -constraint problem. Then define $\epsilon = f(\mathbf{x}^*)$, and there exists $\mathbf{x} \in \mathcal{S}$ with $f_i(\mathbf{x}) < f_i(\mathbf{x}^*)$ and $f_j(\mathbf{x}) \leq \epsilon_j = f_j(\mathbf{x}^*)$, which contradicts the Pareto-optimality of $\mathbf{x}^*$.

“ $\Leftarrow$ ” Suppose $\mathbf{x}^*$ is not Pareto-optimal. Then there exists $\mathbf{x} \in \mathcal{S}$, such that $f_i(\mathbf{x}) < f_i(\mathbf{x}^*)$ for at least one $i \in \{1, \dots, m\}$ and $f_j(\mathbf{x}) \leq f_j(\mathbf{x}^*)$ for all $j = 1, \dots, m$. Since $\mathbf{x}^*$ is a solution of the ϵ -constraint problem, it follows that $f_j(\mathbf{x}) \leq f_j(\mathbf{x}^*) \leq \epsilon_j$, so $\mathbf{x}$ satisfies the constraints, which contradicts the optimality of $\mathbf{x}^*$ for the ϵ -constraint problem and index i .

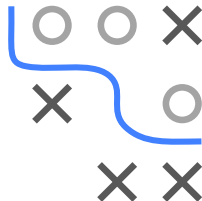
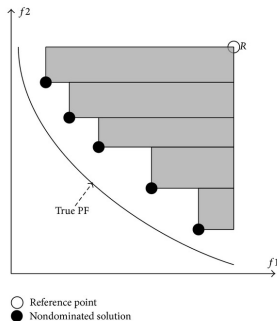


EVALUATION OF SOLUTIONS

A common measure for the performance of a set of candidates $\mathcal{P} \subset \mathcal{S}$ is the **dominated hypervolume**:

$$S(\mathcal{P}, R) = \lambda\left(\bigcup_{\tilde{\mathbf{x}} \in \mathcal{P}} \{\mathbf{x} \mid \tilde{\mathbf{x}} \prec \mathbf{x} \prec R\}\right),$$

where $\lambda(\cdot)$ is the Lebesgue measure and R a pre-specified reference point that every point in the set must dominate.



- HV is measured w.r.t. the reference point R , often chosen so each component is a known “worst” (upper bound) for that objective.
- HV is also called the **S-metric**.
- Its computation is $\mathcal{O}(n^{m-1})$ in the number of points and dimension m .
- Efficient algorithms exist for small m . In ML, we rarely exceed $m = 3$ in practice.